

Final AI Engineer Path (20/80 Rule) – 2026

Goal: Become job-ready for AI Engineer roles by late 2026 or early 2027. Study commitment: ~3–4 hours per day. Strategy: Focus only on the highest-impact AI skills using the 20/80 rule. This roadmap prioritizes practical AI systems, LLM applications, and real product development.

Daily Routine (Throughout the Journey)

- 1 hour: Data Structures & Algorithms practice (LeetCode recommended).
- 3 hours: Learning AI topics and building projects.
- Push code to GitHub regularly.
- Deploy at least two projects publicly.

Stage 1 – Machine Learning Basics (3 Weeks)

Topics:

- NumPy and Pandas fundamentals
- Data preprocessing and feature engineering
- Linear regression
- Logistic regression
- Decision trees
- Model evaluation and overfitting concepts

Project 1 – ML Dataset Analyzer

- Load and analyze datasets
- Generate statistics and feature insights
- Visualize distributions
- Train basic ML models

Stage 2 – Deep Learning Basics (2 Weeks)

- Neural networks
- Activation functions
- Backpropagation
- Convolutional Neural Networks (CNN)
- Embeddings concept

Project 2 – Image Classifier (CNN)

- Build an animal image classifier
- Use PyTorch
- Train CNN model
- Evaluate classification accuracy

Stage 3 – LLM Engineering (4 Weeks)

- Prompt engineering
- Tokens and context windows
- Embeddings
- Vector search
- Context management

Project 3 – AI Chatbot

- Conversation memory
- Prompt templates
- Context aware responses

Project 4 – PDF RAG Chatbot

- Upload PDF documents
- Generate embeddings
- Store vectors
- Answer user questions using retrieval augmented generation

Stage 4 – AI Agents (3 Weeks)

- Autonomous agents
- Tool usage
- Task planning
- Multi-agent collaboration

Project 5 – AI Research Agent

- User requests research topic
- Agent searches internet
- Extracts information
- Generates structured summary

Stage 5 – AI Product Development (3 Weeks)

- Building APIs for AI models
- Backend development
- Creating AI web applications
- Deployment of AI systems

Project 6 – AI Customer Support Bot

- Answer company FAQ
- Retrieve information from knowledge base
- Maintain conversation context
- Deploy as web application

AI Engineer Killer Project

This project is highly impressive for recruiters because it combines RAG, AI agents, APIs, and real product functionality.

Project – Autonomous AI Business Analyst

- User asks for market analysis (example: 'Analyze the electric vehicle market').
- Research agent gathers data from the web.
- Analysis agent interprets market trends.
- Writer agent generates a professional report.
- System exports final business analysis document.

Recommended Final Portfolio (Target)

- ml-dataset-analyzer
- image-classifier-cnn
- ai-chatbot
- pdf-rag-chatbot
- ai-research-agent
- ai-customer-support-bot
- autonomous-ai-business-analyst (killer project)

Preparation Do's

- Code every day.
- Finish projects before starting new ones.
- Write clear README documentation.
- Push code to GitHub frequently.
- Deploy at least two projects online.

Preparation Don'ts

- Do not stay stuck in tutorial loops.
- Do not focus only on theory.
- Do not copy code without understanding.
- Do not ignore deployment and system design.
- Do not start many projects without finishing them.

Following this focused roadmap consistently for 6–8 months can make you competitive for AI Engineer roles. Focus on building working AI systems and maintaining a strong GitHub portfolio.